



If f is continuous on $[a, b]$, then f has an antiderivative on $[a, b]$. Moreover, if F is any antiderivative of f on $[a,$

Answer

☒ $\int_a^b f(x) dx = 0$

☐ $\int_a^b f(x) dx = F(x) - F(a)$

☐ $\int_a^b f(x) dx = F(b) - F(a)$

$\int_a^b f(x) dx = F(b) - F(a)$

Question No : 39 of 50

More: 1 (Time 1 min) !

By putting $x = \sin t$, $\ln 1 - 2x^2 = \underline{\hspace{2cm}}$

Answer

☐ $\sin 2t$ ☐ $-\cos 2t$ ☐ $\cos t$ ☒ $\cos 2t$

06:47

Question No : 40 of 50 Mortx:1 (Time 1 Min)

If f is a differentiable function at a local extreme point x_0 then $f'(x_0) =$

0 Page 119

☐ positive.

☐ negative.

☐ does not exist.

Identify the anti - derivative of the function $f(x) = x^2 + \sin x$

Answer

☐ $F(x) = \frac{x^2}{2} - \sin x.$

☒ $F(x) = \frac{x^3}{3} - \cos x.$

☐ $F(x) = \frac{x^4}{4} - \cos x.$

☐ $F(x) = \frac{x^3}{3} + \cos x.$

Question No : 36 of 50

For the function $F(x) = \int_0^x f(t) dt = \frac{x^2}{2}, 0 < x \leq 1$.

Identify the function such that $F(x)$ is derivative of the function

☐ $f(x) = 2x, 0 \leq x \leq 1$

☒ $f(x) = x, 0 \leq x \leq 1$

☐ $f(x) = x^2 + 1$

☐ $f(x) = x - 2, 0 \leq x \leq 1$

Question No : 35 of 50

MATHS 1 (12TH CLASS)

Lower sum of f over P is

☒ $s(P) = \sum_{j=1}^n m_j (x_j - x_{j-1}).$

☐ $\int_a^b f(x) dx.$

☐ $\int_a^b f(x) dx.$

☐ $\sum_{j=1}^n m_j (x_j - x_{j-1}).$



If f and g are differentiable on an interval, and if $f'(x) = g'(x)$ for all x in that interval, then

05:32

Answer

- ☒ fg is a constant function on the interval.
- ☐ $\frac{f}{g}$ is a constant function on the interval.
- ☐ $f - g$ is a constant function on the interval.
- ☐ $f + g$ is a constant function on the interval.



$\lim_{x \rightarrow 0} (\cot x)^{\sin 2x}$ has the following indeterminate form _____.

Answer



1^∞



0^∞



0^0



∞^0





Question No : 31 of 50

Marks : 1 (Time 1 Min)

If $m \leq f(x) \leq M$ for all x in $[a, b]$, then

Answer

- ☐ $M(b-a) \leq s(P) \leq S(P) \leq m(b-a)$ for every partition P .
- ☒ $m(b-a) \leq s(P) \leq S(P) \leq M(b-a)$ for every partition P .
- ☐ $m(b-a) \leq S(P) \leq s(P) \leq M(b-a)$ for every partition P .
- ☐ $m(b+a) \leq S(P) \leq s(P) \leq M(b+a)$ for every partition P .

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08:10



Question No : 24 of 50

Marks : (Time 1 min)

If $f(x) = e^x$, then $f''(x) = \underline{\hspace{2cm}}$.



Answer

☐ e ☐ e^{x^2} ☒ e^x ☐ ne^x

$\lim_{x \rightarrow 0^+} x \log x$ has the following indeterminate form _____.

Answer

$(0)(\infty)$

☐

0

☐

0^0

☐

$(\infty)(\infty)$

Question No : 28 of 50

Marked (Time 1 min) 1

The function $f(x) = x$ over the interval $[2,3]$ is



Answer

☐ Riemann integrable function.☒ Riemann integrable and equals to $\frac{5}{2}$.☐ Riemann integrable and equals to $\frac{2}{5}$.☐ is not Riemann integrable.

Question No: 29 of 50

Marks: 1 (Time: 1 min)

If f is unbounded on the finite closed interval $[a, b]$ then



Answer

- ☐ f must be continuous.
- ☐ f must have extreme values inside the interval $[a, b]$.
- ☐ f is Riemann integrable.
- ☒ f is not Riemann integrable.

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Upper sum of f over P

☒ $S(P) = \sum_{j=1}^n M_j (x_j - x_{j-1}).$

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☐ $S(P) = \sum_{j=1}^n m_j (x_j - x_{j-1}).$

☐ $\int_a^b f(x) dx.$

☐ $\int_a^b f(x) dx.$

For the function $f(x) = x$, $a \leq x \leq b$, Riemann sums are of the form

87.65

Answer

☐ $\sigma = \sum_{j=1}^n (x_j - x_{j-1}).$

PAGE 136

☒ $\sigma = \sum_{j=1}^n c_j (x_j - x_{j-1}).$

☐ $\sigma = \sum_{j=1}^n c_j.$

Question No : 16 of 50

11

Marks: 1 (Time 1 Min)

If $f(x) = \log x$ and $g(x) = \frac{1}{1-x^2}$, then $g \circ f$ is

Answer

☐ $\log \frac{1}{1-x^2}$

☒ $\frac{1}{1-(\log x)^2}$

☐ $\log \frac{1}{x^2}$

6426

Question No: 17 of 50

More: (Time 1 min)

The function $h(x) = -x^3$ is

04:32

Answer

- ☐ undefined on $(-\infty, \infty)$.
- ☒ decreasing on $(-\infty, \infty)$.
- ☐ increasing on $(-\infty, \infty)$.
- ☐ positive on $(-\infty, \infty)$.

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Generalized Mean Value Theorem says,

If f and g are continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) then

Answer

- ☐ $f'(c) = g'(c)$ for some c in (a, b) .
- ☐ $[g(b) - g(a)] = [f(b) - f(a)]$ for some c in (a, b) .
- ☒ $[g(b) - g(a)] f'(c) = [f(b) - f(a)] g'(c)$ for some c in (a, b) .
- ☐ $f'(c) = g'(c) = 0$ for some c in (a, b) .

The inverse of the function $f(x) = x^2$ is

Answer

☐ $f^{-1}(y) = y^2$

☒ $f^{-1}(y) = \sqrt{y}$

☐ $f^{-1}(y) = \frac{1}{x^2}$

☐ $f^{-1}(y) = \frac{1}{y^2}$

54.48



Question No : 22 of 50

Marks: 1 (Time : Min)

If f is integrable on $[a, b]$ and $a \leq c \leq b$, then the function F defined by $F(x) = \int_c^x f(t)dt$, then which statement is true?

Answer

☐ $F(x)$ is a differentiable function on $[a, b]$.☐ $F(x)$ is a constant function on $[a, b]$.☐ $F(x)$ is a monotonic function on $[a, b]$.☒ $F(x)$ is a Lipschitz continuous on $[a, b]$.

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Question No : 27 of 50

Marks : 1 (Time 1 min)

If P and P' are partitions of $[a, b]$, then P' is a refinement of P

06:43

Answer

☐ P has exactly four points.☐ If every partition point of P' is also a partition point of P .☒ If every partition point of P is also a partition point of P' .☐ None of the given choices.

If a function f is Riemann integrable on $[a,b]$, then it is

Its topic on page 136

06/26

Answer

- ☐ Differentiable on $[a,b]$.
- ☐ None of the given choices.
- ☐ Riemann integrable on the any subinterval of $[a,b]$.
- ☒ Lipschitz continuous on $[a,b]$.

Not sure

Raabe's test says suppose that $a_n > 0$ for large n .

Let $M = \lim_{n \rightarrow \infty} \sup n \left(\frac{a_{n+1}}{a_n} - 1 \right)$ then

Answer

☐ $\sum a_n < \infty$ if $M = -1$

☐ $\sum a_n < \infty$ if $M > -1$

☐ $\sum a_n < \infty$ if $M = 1$

☒ $\sum a_n < \infty$ if $M < -1$

If $\lim_{x \rightarrow 0} f(x) = 10$ and $\lim_{x \rightarrow 0} g(x) = -2$ then $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)}$ is

00:24

Answer



-10



5



0



-5

Question No: 15 of 50

Marked 1 (Time 1 min)

If $f(x) = \log x$ and $g(x) = \frac{1}{1-x^2}$ then $f \circ g$ is

Answer

☒ $(f \circ g)(x) = \log \frac{1}{1-x^2}$

☐ $(f \circ g)(x) = \log \frac{x}{1-x^2}$

☐ $(f \circ g)(x) = \log \frac{1}{x^2}$

-1



For the function $f(x) = \begin{cases} x & 0 \leq x < 1 \\ 2 & 1 \leq x \leq 2 \end{cases}$ which statement is true?

Answer

☐ Continuous at $x = 1$

☐ $\lim_{x \rightarrow 1^+} f(x) = 1$

☐ $\lim_{x \rightarrow 1^-} f(x) = 2$

01:52

the function is nondecreasing on the interval $[0, 2]$.



The value of $x + \lim_{x \rightarrow 0^+} \frac{|x|}{x} =$

Answer

08:37

☐ $x - 1$

☒ $x + 1$

☐ x

☐ $x + \frac{1}{x}$

If f is differentiable at a local extreme point $x_0 \in D_f$, then

94.53

Answer

☐ $f'(x_0) \neq 0.$ ☒ $f'(x_0) = 0.$

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☐ $x_0 = 0.$ ☐ $|f'(x_0)| < 0.$

Question No : 20 of 50

Marks: 1 (Time 1 Min)

If a function f is continuous on the closed interval then f attains its

☒ extreme values in the closed interval.

Page 119

☐ extreme values at one point in the closed interval.

☐ may or may not attain extreme values in the closed interval.

☐ derivative in the closed interval.

Question No: 2 of 50

Module 1 (Topic 1.1.1)

Which of the functions is not uniformly continuous on $(0, 1)$?

Answer

3 / 41

⋮

☐ x^2 ☒ $\frac{1}{x^2}$ ☐

None

$\sum_{k=1}^{\infty} (-1)^k a_k$ is an example of

Answer

☒ power series.

7 / 41

☐ alternating series.



Page 71

☐ increasing sequence .

☐ nonnegative terms.

02:36



A series is said to be conditionally convergent if it converges but does not —

☒ converge absolutely

Page 78

☐ diverge

☐ diverge absolutely

☐ none of these

Question No : 6 of 50

Marks: 1 (Time 1 Min)

Is continuity implies differentiability?

Answer

☒ Yes

Page 114

☐ No☐ Vice Versa☐ None of these

12.00



Question No : 3 of 50 12 Marks : 1 (Time 1 Min)

The function defined by $f(x) = [x]$ has not a derivative at ———

Answer

☒ $x = 0$

☐ $x = 1$

☐ $x = -1$

☐ $x = 2$

Question No: 5 of 50

012

Mark: 1 (Time 1 Min)

$f(x_0)$ is a local maximum value of f if

max

Answer

☐ $f(x) \leq f(x_0)$



Page 118

$\leq$

☒ $f(x) \geq f(x_0)$

☐ $f(x) > f(x_0)$

☐ None of these

Question No : 10 of 50

Mark: 1 (Time 1 Min)

If $f(x) = 1 - \frac{1}{x^2}$, then $\lim_{x \rightarrow \infty} f(x) = \underline{\hspace{2cm}}$.

$$1 - \frac{1}{x^2}$$

Answer

☐ -1

☐ ∞

☒ 1

☐ 0

Question No : 1 of 50

Marks: 1 (Tim

$$\sin 2t \Big|_{-\pi/4}^{\pi/4} =$$

2 / 41



Answer



1



2



4

Question No. 8 of 30

The special case of Taylor series
when $x_0 = 0$ the series is known

02:26

Answer

- ☐ Power series.
- ☒ Maclaurin series.
- ☐ Finite series .
- ☐ series of nonnegative terms.

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Question No: 32 of 50

Marks: 1 (Time: 1 min)

If f and g are integrable functions on $[a, b]$ then

08:23

Answer

☐ $\sqrt{f}$ is also integrable on $[a, b]$.

☒ fg is anti – derivative of f and g on $[a, b]$.

☐ f^2g is also integrable on $[a, b]$.

☐ None of the given choices.

$$\lim_{x \rightarrow 0} \frac{\sin x}{\tan x} =$$

Answer



0

4 / 41



2



1

01:33



infinity

Question No : 41 of 50

Marks: 3 (Time 6 Min) !

For the power series $\sum \frac{\sin \frac{n\pi}{6}}{2^n} (x-1)^n$ find R

10:11

Answer

← → | **B** *I* U | ✓ $\mathcal{I}_x$ | $\frac{1}{x}$ $\frac{1}{x^2}$ | $\frac{1}{x^3}$ $\frac{1}{x^4}$ | $\frac{1}{x^5}$ $\frac{1}{x^6}$ | $\frac{1}{x^7}$ $\frac{1}{x^8}$ | $\frac{1}{x^9}$ $\frac{1}{x^{10}}$ | $\frac{1}{x^{11}}$ $\frac{1}{x^{12}}$ | Styles | Format | Font | Size |

A - **A** - $\mathcal{I}_x$ |

Solution: For R , we have

$$\begin{aligned}\limsup_{n \rightarrow \infty} |a_n|^{1/n} &= \limsup_{n \rightarrow \infty} \left(\frac{|\sin n\pi/6|}{2^n} \right)^{1/n} \\ &= \frac{1}{2} \limsup_{n \rightarrow \infty} (|\sin n\pi/6|)^{1/n} \\ &= \frac{1}{2}(1) = \frac{1}{2}.\end{aligned}$$

Therefore, $R = 2$ and Theorem about the convergence of power series implies that the series converges absolutely uniformly in closed subintervals of $(-1, 3)$ and diverges if $x < -1$ or $x > 3$.

Let $f(x) = \begin{cases} 0 & \text{if } x \text{ is irrational,} \\ 1 & \text{if } x \text{ is rational,} \end{cases}$

Find $\int_a^b f(x) dx$.

Answer



✓
Example: Let

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is irrational,} \\ 1 & \text{if } x \text{ is rational,} \end{cases}$$

Let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$. Since every interval contains both rational and irrational numbers

$$m_j = 0 \quad \text{and} \quad M_j = 1, \quad 1 \leq j \leq n.$$

Hence,

$$S(P) = \sum_{j=1}^n 1 \cdot (x_j - x_{j-1}) = b - a$$

$$s(P) = \sum_{j=1}^n 0 \cdot (x_j - x_{j-1}) = 0.$$

Since all upper sums equal $b - a$ and all lower sums equal 0, from definition we have

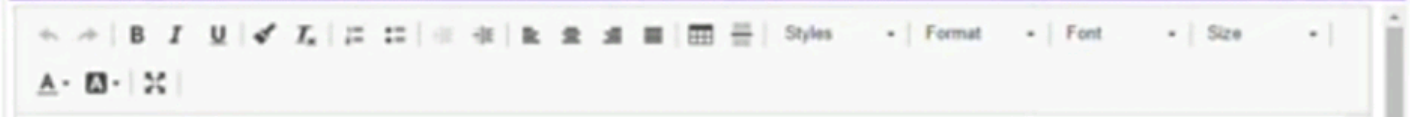
$$\overline{\int_a^b} f(x) dx = b - a \quad \text{and} \quad \underline{\int_a^b} f(x) dx = 0.$$

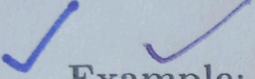
Question No : 43 of 50 9 Marks: 3 (Time 6 Min)

Check the uniform continuity of the function $f(x) = 2x$.



Answer





Example: Check the uniform continuity of the function

$$f(x) = 2x.$$

Solution: For the function $f(x)$, we have

$$|f(x) - f(x')| = 2|x - x'| < \varepsilon \quad \text{if} \quad |x - x'| < \varepsilon/2.$$

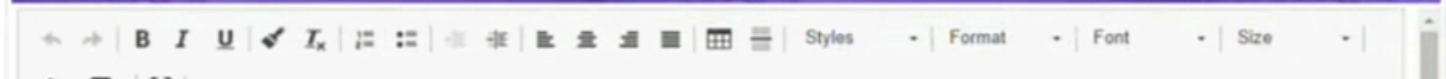
The function $f(x)$ is uniformly continuous on $(-\infty, \infty)$,

Question No : 44 of 50 Marks: 3 (Time 6 Min)

Question No : 44 of 50 9 Marks: 3 (Time 6 Min) !

Calculate the derivative of $h(x) = \sin \frac{1}{x}$, $x \neq 0$ using chain rule.

Answer



✓ **Example:** Calculate the derivative of

$$h(x) = \sin \frac{1}{x}, \quad x \neq 0.$$

If

$$f(x) = \sin x \quad \text{and} \quad g(x) = \frac{1}{x}, \quad x \neq 0,$$

then

$$h(x) = f(g(x)) = \sin \frac{1}{x}, \quad x \neq 0,$$

and

$$\underline{h'(x) = f'(g(x))g'(x) = \left(\cos \frac{1}{x}\right) \left(-\frac{1}{x^2}\right), \quad x \neq 0.}$$

Let's break down the proof step by step:

Given $\epsilon > 0$, we want to find a $\delta > 0$ such that for all x , if $0 < |x - 2| < \delta$, then $|x^2 - 4| < \epsilon$.

First, let's work with the inequality $|x^2 - 4| < \epsilon$:

$$|x^2 - 4| = |x^2 - 2^2| = |(x - 2)(x + 2)| = |x - 2||x + 2|$$

Since we are interested in values of x close to 2, we can assume that $|x - 2| < 1$, which implies that $1 < x < 3$. This also means that $|x + 2| < 5$, because when $x < 3$, $x + 2 < 5$, and when $x > 1$, $x - 2 > -1$ which means $|x + 2| = x + 2$.



Determine whether the function $f(x) = 1 + x$ is integrable on $[a, b]$?

12:13

Answer

Rich text editor toolbar with icons for undo, redo, bold, italic, underline, link, unlink, bulleted list, numbered list, decrease indent, increase indent, table, table border, styles, format, font, and size.

$$\int_a^b f(x) dx = \int_a^b f_1(x) dx + \int_a^b f_2(x) dx$$

$$\int_a^b f(x) dx = \int_a^b 1 dx + \int_a^b x dx$$

$$\int_a^b f(x) dx = b-a + \frac{b^2-a^2}{2}$$

So $f(x) = 1+x$ is
integrable on $[a, b]$

Question No : 47 of 50

Marks: 5 (Time 10 Min)

$$\text{Let } f(x) = \begin{cases} -1, & x \text{ is irrational} \\ 1, & x \text{ is rational} \end{cases}$$

Show that f is not continuous anywhere.

Answer

12:42



As f is a continuous function

For the piecewise defined function $f(x) = \begin{cases} x^3, & x \leq 0 \\ x^2 \sin \frac{1}{x}, & x > 0 \end{cases}$ investigate the one sided derivatives.

Answer

← → | **B** **I** **U** | I_x | $\frac{1}{x}$ $\frac{1}{x^2}$ | $\frac{1}{x^3}$ $\frac{1}{x^4}$ | $\frac{1}{x^5}$ $\frac{1}{x^6}$ | $\frac{1}{x^7}$ $\frac{1}{x^8}$ | $\frac{1}{x^9}$ $\frac{1}{x^{10}}$ | Styles | Format | Font | Size

A - **A** - |

✓ Example: For the piecewise defined function

$$f(x) = \begin{cases} x^3, & x \leq 0, \\ x^2 \sin \frac{1}{x}, & x > 0, \end{cases}$$

Investigate the one sided derivatives.

Solution: For the given function, we have

$$f'(x) = \begin{cases} 3x^2, & x < 0, \\ 2x \sin \frac{1}{x} - \cos \frac{1}{x}, & x > 0. \end{cases}$$

Since neither formula in $f(x)$ holds for all x in any neighborhood of 0, we cannot simply differentiate either to obtain $f'(0)$.

We will calculate the one sided derivatives

$$\lim_{x \rightarrow 0+} f(x) = \lim_{x \rightarrow 0+} \frac{x^2 \sin \frac{1}{x} - 0}{x - 0} = \lim_{x \rightarrow 0+} x \sin \frac{1}{x} = 0,$$

$$f'_-(0) = \lim_{x \rightarrow 0-} \frac{x^3 - 0}{x - 0} = \lim_{x \rightarrow 0-} x^2 = 0;$$

hence, $f'(0) = f'_+(0) = f'_-(0) = 0$.

ided derivatives and one sided limit of

Question No : 49 of 50

Marks: 5 (Time 10 Min)



Suppose that F is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) , and f is integrable on $[a, b]$. Suppose also that $F'(x) = f(x)$, $a < x < b$.

Then prove that $\int_a^b f(x) dx = F(b) - F(a)$.

Answer

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Suppose that f is continuous on the closed interval $[a, b]$ and diff on the open interval (a, b) . and the f is

on $[a, b]$.

✓ **Theorem:** If f is continuous on $[a, b]$, then f has an antiderivative on $[a, b]$.
Moreover, if F is any antiderivative of f on $[a, b]$, then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Proof: The function $F_0(x) = \int_a^x f(t) dt$ is continuous on $[a, b]$.

Furthermore, $F'_0(x) = f(x)$ on (a, b) by previous theorem.

Therefore, F_0 is an antiderivative of f on $[a, b]$.

Now let $F = F_0 + c$ ($c = \text{constant}$) be an arbitrary antiderivative of f on $[a, b]$.

Then

$$F(b) - F(a) = \int_a^b f(x) dx + c - \int_a^a f(x) dx - c = \int_a^b f(x) dx.$$

When applying this theorem, we will use the familiar notation

$$F(b) - F(a) = F(x) \Big|_a^b.$$

Apply Dirichlet's test to determine convergence of the following series

$$\sum_{n=2}^{\infty} \frac{\sin n\theta}{n + (-1)^n} = \frac{13}{31} \pi \quad (k = \text{integer}).$$

Answer



✓ Example: Apply Dirichlet's test to the following series

$$\sum_{n=2}^{\infty} \frac{\sin n\theta}{n + (-1)^n}, \quad \theta \neq k\pi \quad (k = \text{integer}),$$

Solution: We take

$$\underline{a_n} = \frac{1}{n + (-1)^n} \quad \text{and} \quad \underline{b_n} = \sin n\theta.$$

Then $\lim_{n \rightarrow \infty} a_n = 0$, and

$$|a_{n+1} - a_n| < \frac{3}{n(n-1)}$$

(verify), so

$$\sum |a_{n+1} - a_n| < \infty.$$

Now

$$B_n = \sin 2\theta + \sin 3\theta + \cdots + \sin n\theta.$$

2.17. Dirichlet's Test for Series

We can write

$$\begin{aligned} B_n &= \frac{(\cos \frac{3}{2}\theta - \cos \frac{5}{2}\theta) + \cdots + (\cos (n - \frac{1}{2})\theta - \cos (n + \frac{1}{2})\theta)}{2 \sin(\theta/2)} \\ &= \frac{\cos \frac{3}{2}\theta - \cos (n + \frac{1}{2})\theta}{2 \sin(\theta/2)}, \end{aligned}$$

which implies that $|B_n| \leq \left| \frac{1}{\sin(\theta/2)} \right|$, $n \geq 2$.

Since $\{a_n\}$ and $\{b_n\}$ satisfy the hypotheses of Dirichlet's theorem, $\sum a_n b_n$ converges.